ANTONIO JAMES BUTLER

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Education

University of Connecticut August 2020 – May 2024
B.S. in Computer Science, *cum laude* Cumulative GPA: 3.831/4.0
Honors: Babbidge Scholar (2022), Dean's List (2021 – 2022)

Experience

QCDx May 2023 – June 2026
Software Engineer Farmington, CT

- Designed and maintained backend infrastructure for clinical data and high-volume microscopy datasets, enabling reliable, scalable storage and retrieval.
- Engineered GPU-accelerated image processing pipelines in Python using OpenCL, optimizing memory utilization and processing throughput for large multidimensional microscopy datasets.
- Developed desktop applications and automation tools in Java and Python supporting microscopy acquisition, image management, and laboratory workflows.
- Integrated laboratory instrumentation, image acquisition software, and backend systems to automate microscopy imaging workflows.
- Partnered with interdisciplinary scientists and engineers to translate research requirements into production software supporting clinical imaging workflows.
- Investigated and resolved complex issues across image acquisition, backend systems, and image processing pipelines, improving software reliability.

University of Connecticut – School of Engineering January 2023 – May 2023
Undergraduate Teaching Assistant Storrs, CT

- Mentored undergraduate computer science students through laboratory instruction, debugging sessions, and code reviews, reinforcing students' problem-solving and software development skills.

Barnes and Noble Education August 2022 – December 2022
Technology Sales Associate Storrs, CT

- Provided technical sales support for consumer electronics while assisting with hardware diagnostics, repairs, and warranty services.

Regional School District No. 6 July 2022 – August 2022
IT Technician Litchfield, CT

- Developed and implemented an efficient software deployment process for student laptops, streamlining operating system and application installation.
- Configured, deployed, and troubleshot district technology—including MacBooks, Smart Boards, and point-of-sale systems—to support school-wide operations.

Technical Strengths

Languages	Python, Java, C++, C, SQL, JavaScript
Frameworks & Libraries	Swing, JavaFX, OpenCL, NumPy, Pandas
Databases	SQL Server
Developer Tools	Maven, Conda, Git
Technologies	GPU Computing, Image Processing, Scientific Computing

Leadership & Projects

Senior Design Project: Secure Embedded Architecture September 2023 – May 2024
Team Lead

- Led a team implementing the CRYSTALS-Dilithium post-quantum cryptographic algorithm in the First Stage Boot Loader (FSBL) of a Xilinx ZCU102 FPGA platform, designing a secure boot authentication protocol and integrating signature verification with hardware-backed key storage.

Husky Developers September 2022 – September 2023
Project Lead

- Led development of a 3D website for the organization using Three.js and Blender, independently learning new technologies while mentoring fellow club members.

Hack UConn March 4, 2023
Second Place

- Awarded second place for developing an AI-powered interactive storytelling application integrating Dialogflow, OpenAI's API, and DALL·E image generation.